FUNDAMENTAL OF AI&ML PROJECT REPORT (CSA2001)

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1. Problem Statement

In daily student life maintaining consistency in productive habits such as studying, exercising and managing time is a major challenge. Most students start with motivation but fail to sustain habits due to lack of tracking, feedback and insights.

Existing solutions are often either -

Too complex
Not personalized
Or lacking intelligent insights

This creates a gap for a simple yet intelligent system that not only tracks habits but also predicts behavior and provides insights.

2. Why This Problem Matters

This problem is important because -

- Poor habits directly affect academic performance
- Lack of discipline impacts mental and physical health
- Students are often unaware of their behavior patterns
- There is a need for data-driven self-improvement tools

By solving this problem we help students -

- Build consistency
- Improve productivity
- Make informed decisions about their routines

3. Objective

The objectives of this project are -

- To design a system that tracks daily habits
- To analyze user behavior over time
- To apply machine learning techniques for prediction
- To provide insights that improve consistency
- To demonstrate real-world application of AIML concepts

4. Proposed Solution

The proposed solution is an AI-based Daily Habit Tracker that -

> Allows users to add and track habits daily

> Maintains a history of user activity

> Calculates streaks and performance

> Uses machine learning (Logistic Regression)to -

- Predict whether a habit will be completed
- Provide confidence scores

> Generates basic analytics for improvement

5. Methodology

The system is implemented in the following stages -

5.1 Data Collection

> User inputs daily habit completion (Yes/No)
> Data is stored in a JSON file
> Each record includes -

- Date
- Habit status (1 or 0)

5.2 Data Preprocessing

> Dates are converted into numerical features (day of week)
> Example

- Monday → 0
- Tuesday → 1

5.3 Model Selection

> We use the machine learning algorithm -
> Logistic Regression

Why Logistic Regression?

- Suitable for binary classification (Done / Not Done)
- Simple and efficient
- Works well with small datasets

5.4 Model Training

> Input (X) - Day of the week

> Output (y) - Habit completion (0 or 1)

> Model learns patterns from past data

5.5 Prediction

> Predicts probability of completing a habit on current day

> Output includes -

- Prediction (Yes/No)
- Confidence score

5.6 Evaluation

> Accuracy improves with more data

> System becomes more personalized over time

6. Key Features

> Habit creation and tracking
> Daily check-in system
> Streak tracking for motivation
> Progress visualization
> Reminder system
> Basic analytics
> AI-based prediction system

7. Key Decisions Made

1. Choosing a Real-Life Problem

Selected a problem directly relevant to student life for practical impact.

2. Keeping the System Simple

Used a console based interface to focus on logic rather than UI complexity.

3. Selecting Logistic Regression

Chose a simple ML model to ensure -

- Easy implementation
- Clear understanding
- Reliable results

4. Feature Selection

Used day of the week as the main feature to -

- Identify behavioral patterns
- Keep the model lightweight

8. Challenges Faced

1. Limited Data
   - Machine learning requires sufficient data
   - Initial predictions may not be accurate
2. Feature Simplicity
   - Using only day of week limits prediction accuracy
3. Model Integration
   - Combining ML with a basic Python system required careful design
4. User Input Dependency
   - Accuracy depends on consistent user input

9. Results and Observations

> The system successfully tracks habits and maintains history
> The ML model can -
- Identify patterns
- Predict habit completion with reasonable confidence

> Predictions improve as more data is collected

10. Limitations

> Uses limited features for prediction
> No graphical interface
> No real-time notifications
> Works only on local system
> Requires manual data entry

11. What I Learned

Technical Skills

- Python programming
- Data handling using JSON
- Machine learning using Logistic Regression
- Model training and prediction

Conceptual Understanding

- Importance of data in AI systems
- Feature selection and its impact
- Real-world application of AIML

Practical Insights

- Even simple ML models can solve real problems
- Data consistency is crucial
- AI systems improve over time with usage

12. Conclusion

The AI-Based Daily Habit Tracker successfully addresses the problem of inconsistency in student habits. It combines basic programming with machine learning to create a meaningful and practical solution.
This project demonstrates how AIML concepts can be applied to reallife scenarios in a simple yet effective manner. With further enhancements it has the potential to become a powerful productivity tool.

13. References

Python Documentation
Scikit learn Documentation
Course Materials (AIML)